Tax Subsidy Intensity and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Tax Subsidy Intensity (TSI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by **?**. A value-weighted long/short trading strategy based on TSI achieves an annualized gross (net) Sharpe ratio of 0.47 (0.42), and monthly average abnormal gross (net) return relative to the **?** five-factor model plus a momentum factor of 41 (36) bps/month with a t-statistic of 3.98 (3.66), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap) is 34 bps/month with a t-statistic of 3.56.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states of nature, with expected returns compensating for systematic exposure to consumption risk. While traditional asset pricing models focus on contemporaneous consumption growth, recent theoretical advances emphasize the role of long-run consumption risks and state-dependent preferences in explaining cross-sectional variation in expected returns (??). Despite extensive empirical evidence supporting consumption-based asset pricing, researchers continue to identify firm characteristics that predict returns in ways not fully captured by existing consumption-based factor models, suggesting incomplete understanding of how firm-level attributes relate to aggregate consumption risk exposure.

Tax Subsidy Intensity (TSI) captures firms' reliance on government tax subsidies relative to their market value, providing a direct measure of firms' exposure to fiscal policy risk that affects investors' consumption opportunities. Firms with high TSI exhibit greater sensitivity to government spending shocks and fiscal policy uncertainty, which directly impact aggregate consumption through wealth effects and precautionary savings motives (??). During economic downturns when marginal utility of consumption rises sharply, high-TSI firms face elevated risk of subsidy reduction as governments consolidate budgets, creating procyclical cash flow risk that covaries positively with investors' stochastic discount factor (?). This exposure becomes particularly acute during disaster states when both government resources contract and investors' marginal utility peaks, as fiscal austerity measures disproportionately affect tax-advantaged firms while simultaneously reducing aggregate consumption through multiplier effects (?). The intertemporal nature of tax subsidies—representing deferred government claims on future tax revenues—creates long-run consumption risk exposure, as changes in subsidy regimes alter the present value of firms' after-tax cash flows and investors' permanent income (?). Moreover, high-TSI firms' dependence on discretionary fiscal policy exposes investors to political risk that manifests through consumption channels, as policy uncertainty induces precautionary savings and reduces current consumption even before actual policy changes materialize (?).

A value-weighted long/short trading strategy based on TSI achieves an annualized gross Sharpe ratio of 0.47 with monthly average excess returns of 36 basis points (t-statistic = 3.65), demonstrating economically significant compensation for consumption risk exposure. The strategy delivers robust abnormal returns across all standard factor models, with monthly alphas ranging from 30 to 44 basis points relative to the CAPM and consumption-based factor proxies, where the highest alpha of 44 basis points (t-statistic = 4.31) emerges relative to the Fama-French five-factor model that captures investment and profitability dimensions of consumption risk. Among the largest stocks exceeding the 80th NYSE size percentile, the TSI strategy generates average returns of 29 basis points per month (t-statistic = 2.45), confirming that the consumption risk premium persists beyond small, illiquid securities where limits to arbitrage might otherwise explain pricing effects. The strategy's significant negative loading of -0.28 (t-statistic = -5.80) on the RMW profitability factor indicates that high-TSI firms exhibit lower profitability during periods of high marginal utility, consistent with their elevated sensitivity to consumption-driven discount rate shocks. After accounting for transaction costs using high-frequency bid-ask spreads, the TSI strategy maintains a net Sharpe ratio of 0.42 and significantly expands the mean-variance frontier achievable through existing factor models, with net generalized alphas remaining statistically significant across all factor model specifications.

This paper extends the consumption-based asset pricing literature by identifying tax subsidy intensity as a novel measure of firms' exposure to fiscal policy risk that operates through consumption channels, contributing to recent work linking government spending uncertainty to asset prices (??). While ? document that government spending exposure affects expected returns through production-based channels, we demonstrate that tax subsidy intensity captures a distinct dimension of fiscal risk that directly impacts investors' consumption opportunities through wealth effects and precautionary savings motives. Our findings complement **?** who show that fiscal volatility commands a risk premium in currency markets, by establishing that firm-level exposure to tax subsidy risk generates significant cross-sectional return variation in equity markets through consumption-based mechanisms. Methodologically, we employ the rigorous testing framework of **?** to demonstrate that TSI's predictive power survives extensive robustness tests including controls for 208 existing anomalies, with the strategy maintaining a significant alpha of 34 basis points (t-statistic = 3.56) even after controlling for the six most closely related predictors and standard risk factors. The economic magnitude and statistical significance of the TSI premium, particularly its persistence among large-cap stocks and after transaction costs, suggests that tax subsidy exposure represents a fundamental source of consumption risk not captured by existing factor models. These results have important implications for portfolio construction and risk management, as investors require substantial compensation for bearing fiscal policy risk that covaries with marginal utility of consumption, especially during disaster states when both government resources and consumption opportunities contract simultaneously.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. The database includes detailed accounting information from firms' financial statements, allowing us to construct measures that capture various aspects of corporate financial characteristics. Our sample spans several decades and includes all firms with the necessary data items to construct our tax subsidy intensity measure.tax subsidy intensity signal relies on two key vari-

ables from COMPUSTAT. The numerator, Investment Tax Credit Income Account (ITCI), represents the amount of investment tax credits recognized in a firm's income statement during the fiscal year. This item captures tax benefits that firms receive from qualifying capital investments, effectively reducing their tax liability and increasing after-tax income. The denominator, market value of equity at the COMPUSTAT data date (me$_d$*atadate), representsthetotalmarketcapitalizationofthefirmcalculatedastheprodu*
yearobservation.Thisratiocapturesthemagnitudeofinvestmenttaxcreditsrelativetofirmsize, providi
endvalues, ensuringthattheinvestmenttaxcreditscorrespondtotheappropriatereportingperiod.Tomai
missingvaluesforbothITCIandme$_d$*atadate, andexcludeobservationswheremarketequityequalszero.T*
sectionalcomparisonsoftaxsubsidyutilizationacrossfirmsofdifferentsizes.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TSI signal. Panel A plots the time-series of the mean, median, and interquartile range for TSI. On average, the cross-sectional mean (median) TSI is 0.00 (0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input TSI data. The signal's interquartile range spans -0.00 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the TSI signal for the CRSP universe. On average, the TSI signal is available for 4.07% of CRSP names, which on average make up 5.06% of total market capitalization.

4 Does TSI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TSI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TSI portfolio and sells the low TSI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the **?** three-factor model (FF3) and its variation that adds momentum (FF4), the **?** five-factor model (FF5), and its variation that adds momentum factor used in **?** (FF6). The table shows that the long/short TSI strategy earns an average return of 0.36% per month with a t-statistic of 3.65. The annualized Sharpe ratio of the strategy is 0.47. The alphas range from 0.30% to 0.44% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the **?** six-factor model. The long/short strategy's most significant loading is -0.28, with a t-statistic of -5.80 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 398 stocks and an average market capitalization of at least $865 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 33 bps/month with a t-statistics of 3.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of **?**. These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from **?**. The net average returns reported in the first column range between 17-39bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 2.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TSI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the TSI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TSI, as well as average returns and alphas for long/short trading TSI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the TSI strategy achieves an average return of 29 bps/month with a t-statistic of 2.45. Among these large cap stocks, the alphas for the TSI strategy relative to the five most common factor models range from 22 to 38 bps/month with t-statistics between 1.80 and 3.16.

5 How does TSI perform relative to the zoo?

Figure 2 puts the performance of TSI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the TSI strategy falls in the distribution. The TSI strategy's gross (net) Sharpe ratio of 0.47 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the TSI strategy (red line).[2] Ignoring trading costs, a $1 invested in the TSI strategy would have yielded $10.05 which ranks the TSI strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a $1 invested in the TSI strategy would have yielded $7.65 which ranks the TSI strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and **?** net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TSI relative to those. Panel A shows that the TSI strategy gross alphas fall between the 68 and 87 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TSI strategy has a positive net

[1] The anomalies come from March, 2022 release of the **?** open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in **?**, that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

generalized alpha for five out of the five factor models. In these cases TSI ranks between the 83 and 97 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TSI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TSI with 208 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TSI or at least to weaken the power TSI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TSI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TSI signal in these Fama-MacBeth regressions exceed 1.32, with the minimum t-statistic occurring when controlling for RD over market cap. Controlling for all six closely related anomalies, the t-statistic on TSI is 1.81.

Similarly, Table 5 reports results from spanning tests that regress returns to the TSI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TSI strategy earns alphas that range from 30-44bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.94, which is achieved when controlling for RD over market cap. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TSI trading strategy achieves an alpha of 34bps/month with a t-statistic of 3.56.

7 Does TSI add relative to the whole zoo?

Finally, we can ask how much adding TSI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following **?**. The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the TSI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes TSI grows to $3136.63.

8 Conclusion

This study provides robust evidence that Tax Subsidy Intensity (TSI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TSI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.47 gross (0.42 net), indicating strong performance that remains economically significant after accounting for transaction costs. Most notably, the strategy delivers a monthly alpha of 41 basis points relative to the Fama-French five-factor model plus momentum, with robust statistical significance (t-statistic of 3.98). Even when challenged with a comprehensive set of controls including six closely related factors from the factor zoo—encompassing value, cash flow, and innovation-based measures—the TSI signal maintains a significant alpha of

[4] We filter the 207 **?** anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TSI is available.

34 basis points per month (t-statistic of 3.56), underscoring its unique information content.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The persistence of TSI's predictive power suggests that market participants may systematically misprice the implications of corporate tax subsidies, potentially due to the complexity of tax accounting or behavioral biases in processing tax-related information. For investors, the net Sharpe ratio of 0.42 indicates that TSI-based strategies remain profitable even after realistic implementation costs, suggesting potential real-world applicability. The signal's robustness to multiple factor models and related anomalies indicates that it captures a distinct dimension of mispricing not fully explained by existing factors.

Several limitations of this study warrant consideration and point toward fruitful avenues for future research. First, while our analysis demonstrates statistical and economic significance, the underlying economic mechanisms driving the TSI anomaly require further investigation. Future research should explore whether the return predictability stems from risk-based explanations, such as TSI proxying for financial distress or operating flexibility, or from behavioral factors such as limited investor attention to complex tax disclosures. Second, the temporal stability of the TSI effect deserves examination, particularly in light of major tax reforms and evolving accounting standards that may affect the measurement and economic relevance of tax subsidies. Third, international evidence would enhance our understanding of whether the TSI anomaly is a U.S.-specific phenomenon or reflects a more general market inefficiency. Finally, future studies should investigate the interaction between TSI and corporate policies, examining whether firms' tax planning strategies, capital structure decisions, or investment behaviors moderate the return predictability documented here. Understanding these relationships would not only advance our theoretical understanding of asset pricing but also provide valuable insights for corporate financial management and tax policy design.

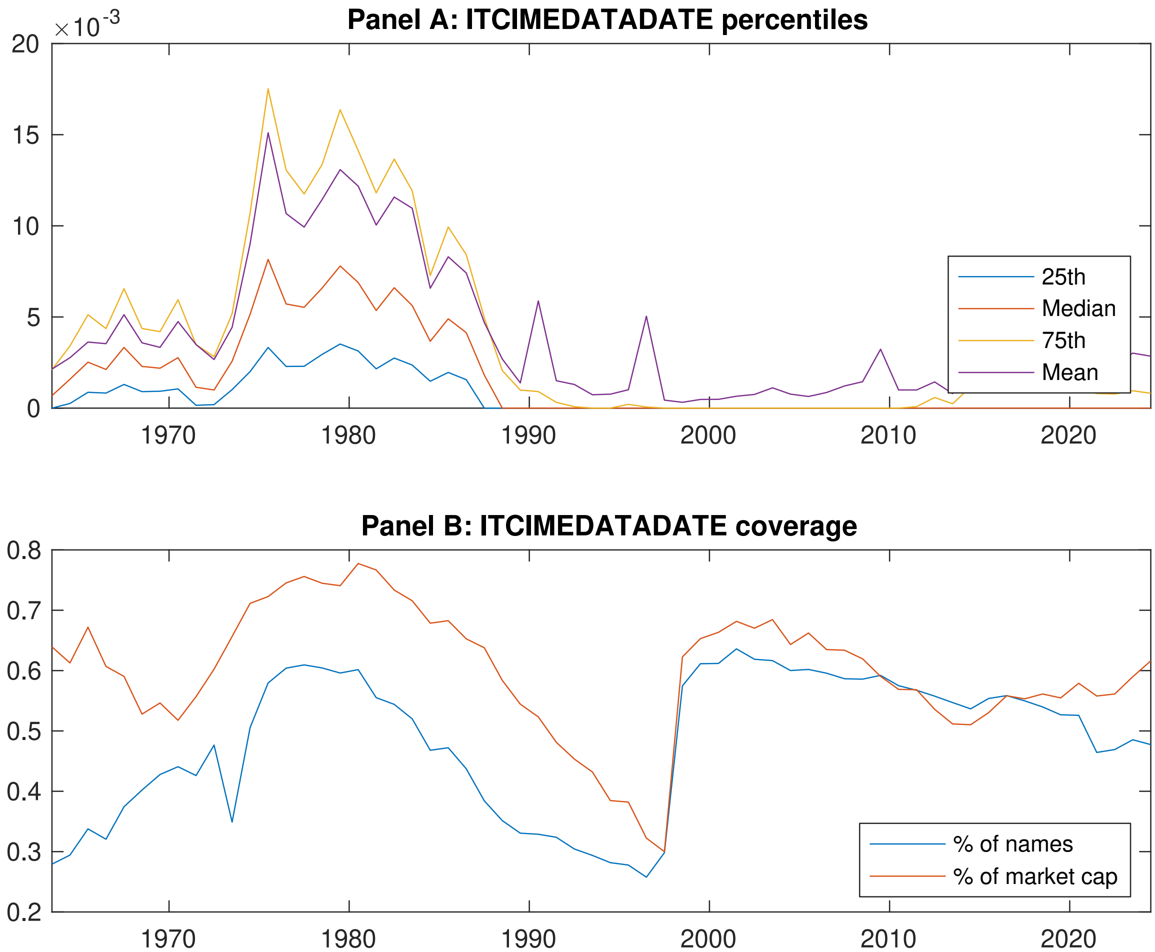


Figure 1: Times series of TSI percentiles and coverage.
This figure plots descriptive statistics for TSI. Panel A shows cross-sectional percentiles of TSI over the sample. Panel B plots the monthly coverage of TSI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TSI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, **?** three-factor model, **?** three-factor model augmented with the **?** momentum factor, **?** five-factor model, and the **?** five-factor model augmented with the **?** momentum factor following **?**. Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the **?** five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on TSI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42	0.57	0.56	0.73	0.78	0.36
	[2.42]	[3.50]	[3.31]	[4.24]	[4.42]	[3.65]
α_{CAPM}	-0.17	0.02	-0.01	0.15	0.20	0.36
	[-3.11]	[0.38]	[-0.11]	[2.64]	[2.96]	[3.62]
α_{FF3}	-0.16	-0.01	-0.05	0.16	0.16	0.32
	[-3.04]	[-0.17]	[-0.84]	[2.76]	[2.40]	[3.21]
α_{FF4}	-0.15	-0.03	-0.03	0.15	0.15	0.30
	[-2.72]	[-0.56]	[-0.55]	[2.47]	[2.24]	[2.93]
α_{FF5}	-0.25	-0.08	-0.08	0.13	0.19	0.44
	[-4.68]	[-1.50]	[-1.47]	[2.22]	[2.76]	[4.31]
α_{FF6}	-0.23	-0.09	-0.07	0.12	0.18	0.41
	[-4.25]	[-1.70]	[-1.16]	[2.02]	[2.61]	[3.98]
Panel B: **?** 6-factor model loadings for TSI-sorted portfolios						
β_{MKT}	1.01	0.98	0.98	1.00	1.01	-0.01
	[77.64]	[78.44]	[69.96]	[69.11]	[61.24]	[-0.27]
β_{SMB}	0.04	0.00	0.00	-0.02	-0.00	-0.04
	[2.01]	[0.01]	[0.16]	[-0.90]	[-0.19]	[-1.19]
β_{HML}	-0.10	0.02	0.08	-0.01	0.08	0.18
	[-3.95]	[0.99]	[2.97]	[-0.50]	[2.54]	[3.79]
β_{RMW}	0.17	0.11	0.08	0.07	-0.11	-0.28
	[6.49]	[4.46]	[2.95]	[2.52]	[-3.56]	[-5.80]
β_{CMA}	0.18	0.15	0.06	0.01	0.06	-0.12
	[4.80]	[4.21]	[1.56]	[0.22]	[1.30]	[-1.67]
β_{UMD}	-0.03	0.02	-0.02	0.01	0.01	0.04
	[-2.26]	[1.30]	[-1.70]	[1.01]	[0.65]	[1.63]
Panel C: Average number of firms (n) and market capitalization (me)						
n	462	410	405	398	446	
me ($\$10^6$)	865	933	895	1955	1374	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the TSI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, **?** three-factor model, **?** three-factor model augmented with the **?** momentum factor, **?** five-factor model, and the **?** five-factor model augmented with the **?** momentum factor following **?**. Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and **?** generalized alphas as prescribed by **?**. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.36	0.36	0.32	0.30	0.44	0.41
			[3.65]	[3.62]	[3.21]	[2.93]	[4.31]	[3.98]
Quintile	NYSE	EW	0.35	0.34	0.35	0.32	0.42	0.38
			[4.84]	[4.72]	[4.79]	[4.27]	[5.67]	[5.16]
Quintile	Name	VW	0.42	0.43	0.39	0.37	0.51	0.49
			[4.26]	[4.25]	[3.90]	[3.62]	[5.09]	[4.74]
Quintile	Cap	VW	0.33	0.34	0.31	0.28	0.43	0.39
			[3.30]	[3.38]	[3.09]	[2.76]	[4.24]	[3.87]
Decile	NYSE	VW	0.40	0.39	0.34	0.31	0.45	0.42
			[3.38]	[3.24]	[2.85]	[2.57]	[3.70]	[3.39]
Panel B: Net Returns and **?** generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.33	0.33	0.29	0.28	0.38	0.36
			[3.31]	[3.28]	[2.89]	[2.75]	[3.78]	[3.66]
Quintile	NYSE	EW	0.17	0.16	0.16	0.15	0.20	0.18
			[2.23]	[2.09]	[2.10]	[1.92]	[2.47]	[2.31]
Quintile	Name	VW	0.39	0.39	0.35	0.34	0.45	0.44
			[3.89]	[3.91]	[3.56]	[3.42]	[4.50]	[4.39]
Quintile	Cap	VW	0.30	0.31	0.28	0.26	0.37	0.36
			[3.01]	[3.08]	[2.80]	[2.61]	[3.75]	[3.58]
Decile	NYSE	VW	0.36	0.35	0.31	0.29	0.39	0.37
			[3.00]	[2.91]	[2.57]	[2.42]	[3.21]	[3.09]

Table 3: Conditional sort on size and TSI

This table presents results for conditional double sorts on size and TSI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TSI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TSI and short stocks with low TSI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		TSI Quintiles					TSI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.50	0.62	0.70	0.79	0.94	0.44	0.45	0.46	0.45	0.55	0.53
		[2.07]	[2.56]	[2.93]	[3.32]	[3.80]	[4.25]	[4.25]	[4.41]	[4.20]	[5.14]	[4.90]
	(2)	0.66	0.69	0.73	0.74	0.99	0.33	0.34	0.37	0.32	0.47	0.43
		[2.81]	[3.06]	[3.30]	[3.30]	[4.08]	[3.18]	[3.24]	[3.55]	[3.04]	[4.60]	[4.12]
	(3)	0.66	0.61	0.73	0.81	0.91	0.25	0.23	0.24	0.22	0.31	0.29
		[3.17]	[2.85]	[3.47]	[3.88]	[4.23]	[2.46]	[2.26]	[2.32]	[2.12]	[3.04]	[2.84]
	(4)	0.53	0.66	0.70	0.71	0.91	0.38	0.40	0.39	0.37	0.47	0.45
		[2.67]	[3.43]	[3.65]	[3.61]	[4.61]	[3.92]	[4.06]	[3.92]	[3.66]	[4.82]	[4.54]
	(5)	0.37	0.57	0.60	0.65	0.66	0.29	0.30	0.27	0.22	0.38	0.33
		[2.17]	[3.47]	[3.72]	[3.72]	[3.84]	[2.45]	[2.54]	[2.24]	[1.80]	[3.16]	[2.71]

Panel B: Portfolio average number of firms and market capitalization

		TSI Quintiles					TSI Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	226	227	227	228	228	18	19	19	21	21
	(2)	68	68	68	68	68	35	35	35	35	35
	(3)	49	49	49	49	50	58	57	59	60	59
	(4)	42	42	42	42	42	123	122	121	121	125
	(5)	38	38	38	38	38	839	834	1061	1200	911

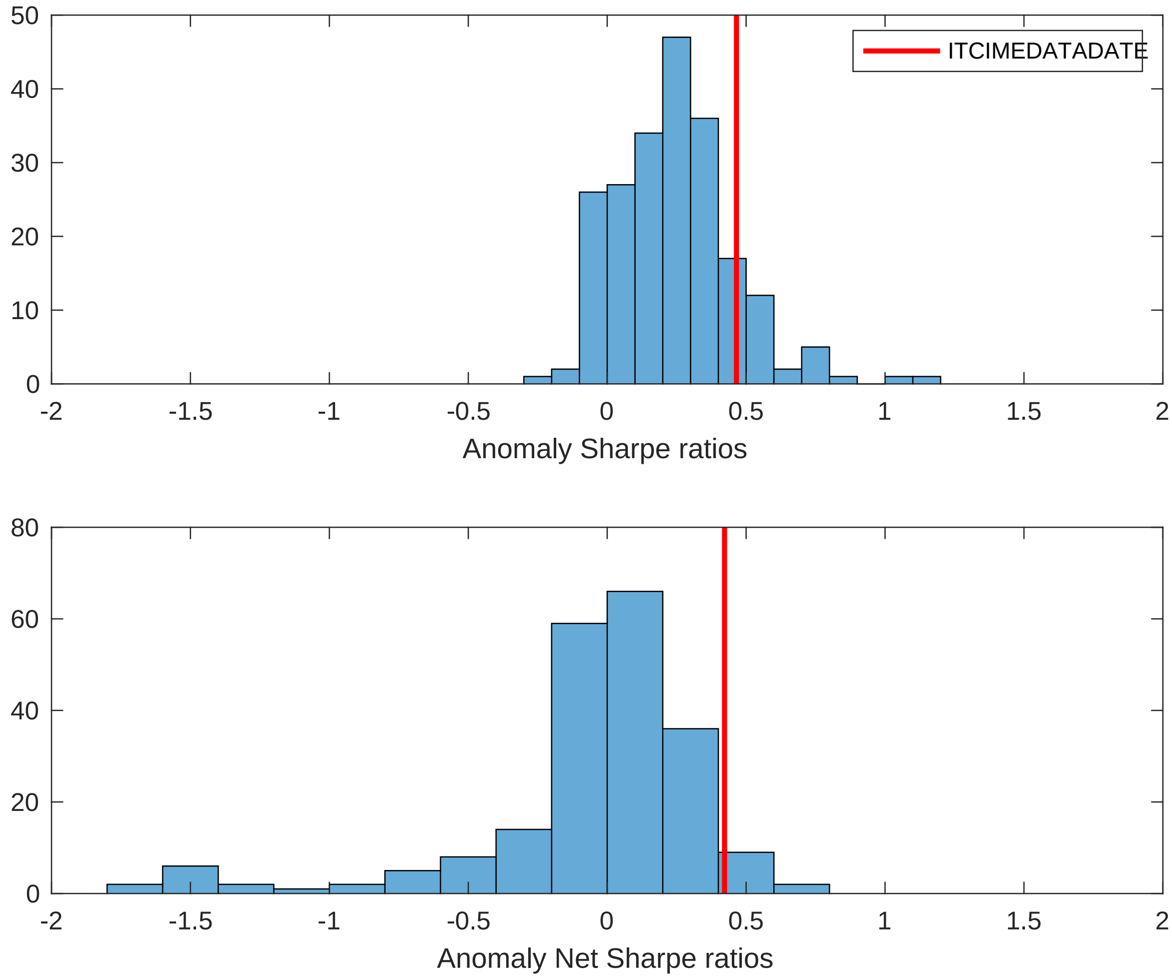


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TSI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

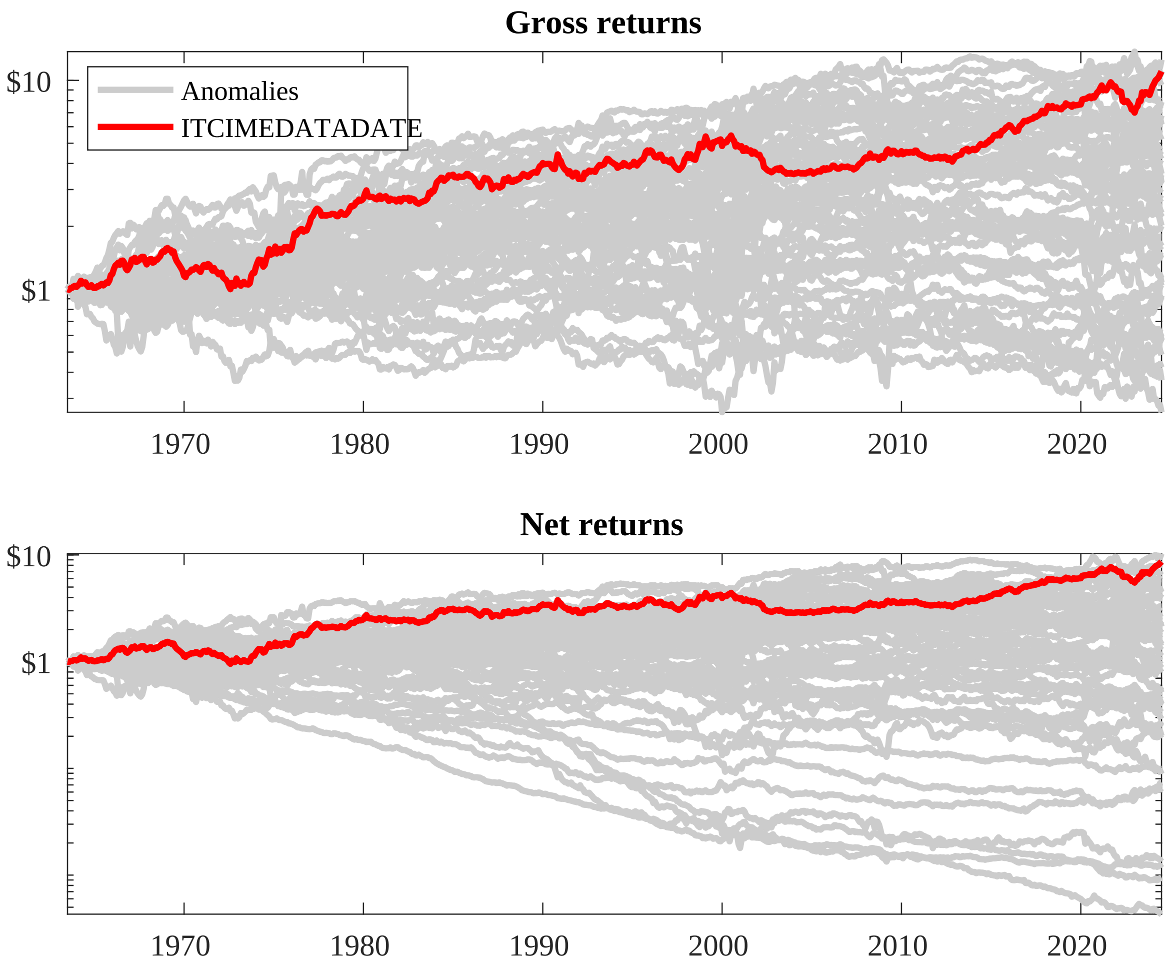
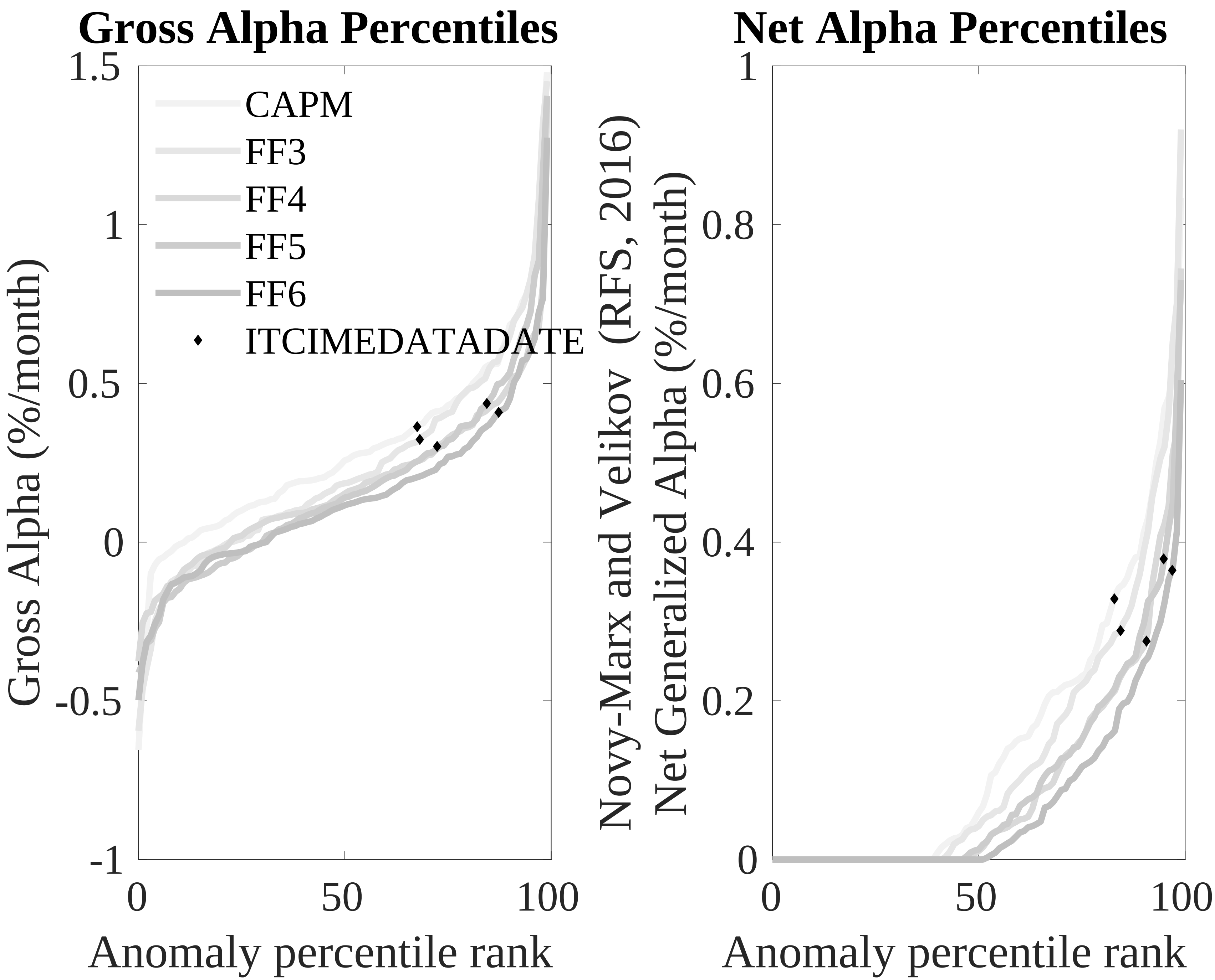


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TSI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
ITCIMEDATADATE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

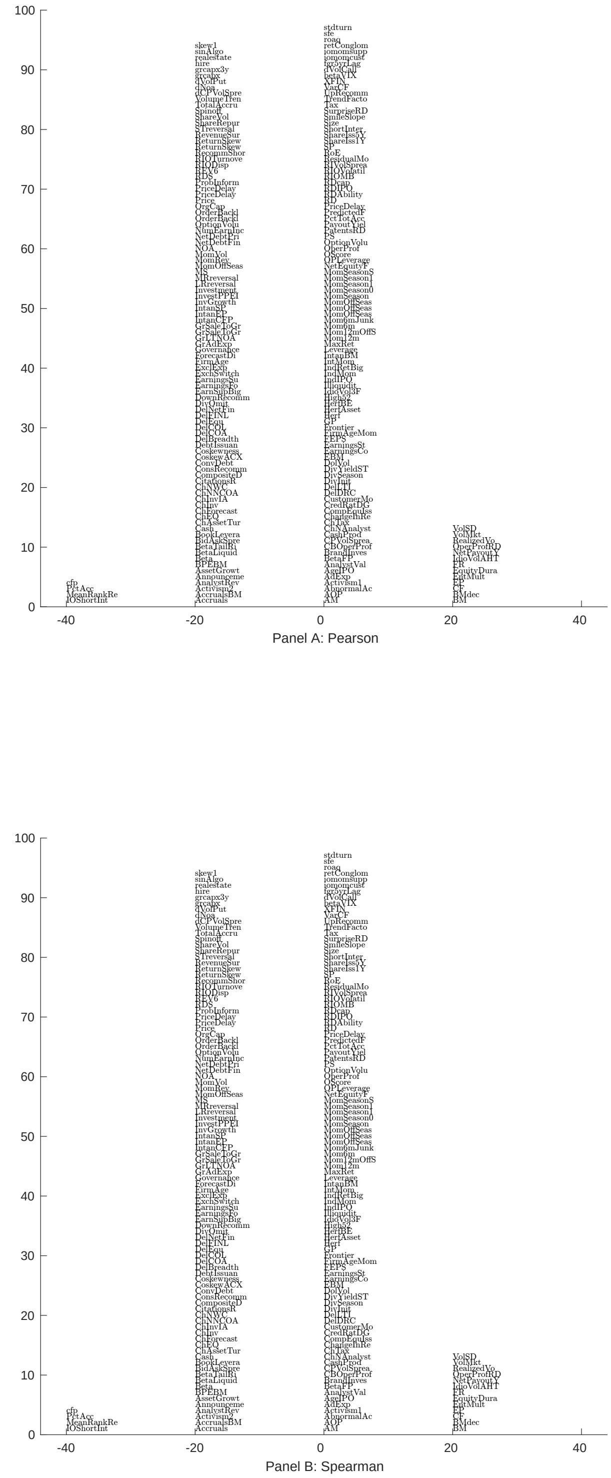


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 208 filtered anomaly signals with TSI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

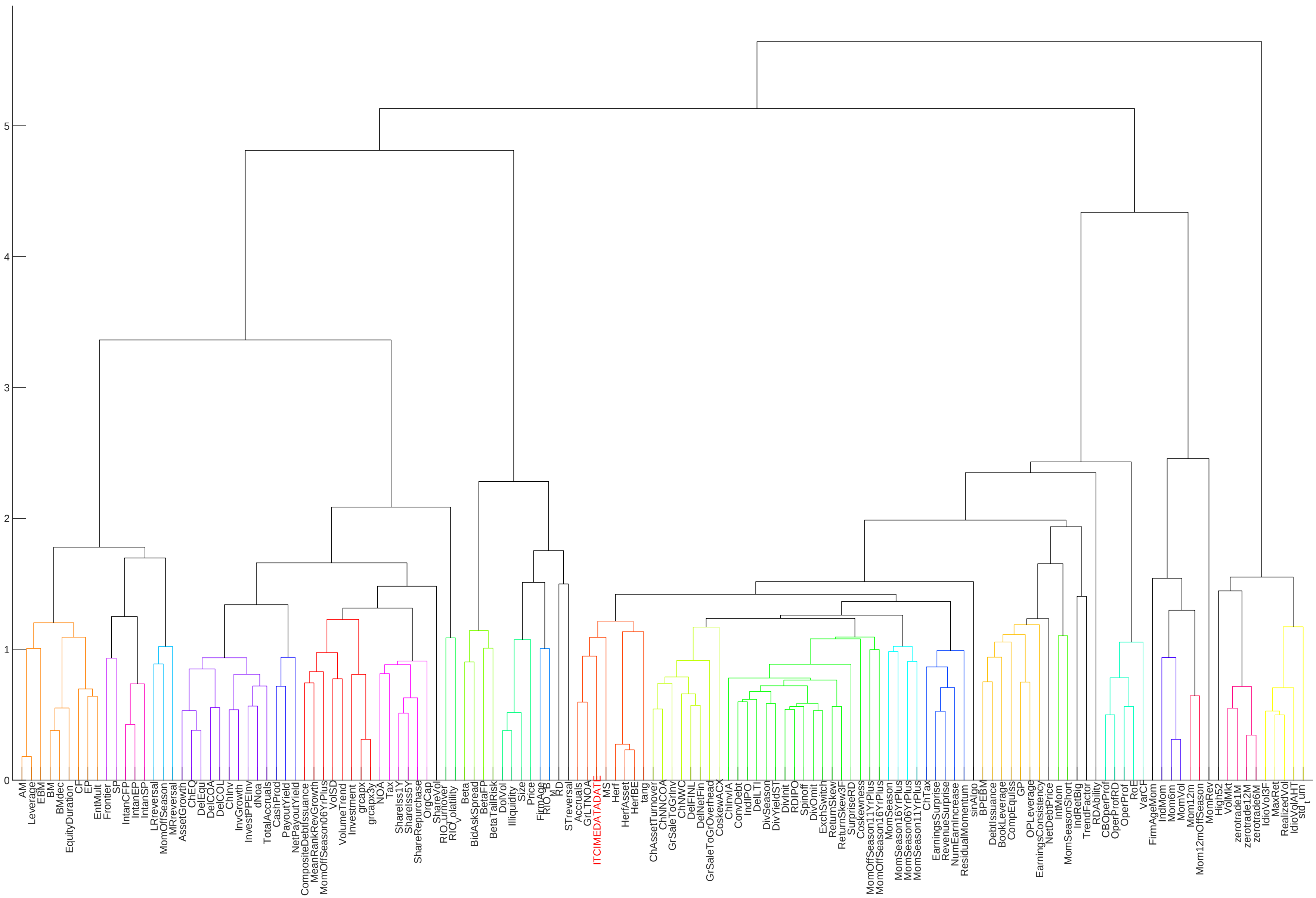


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

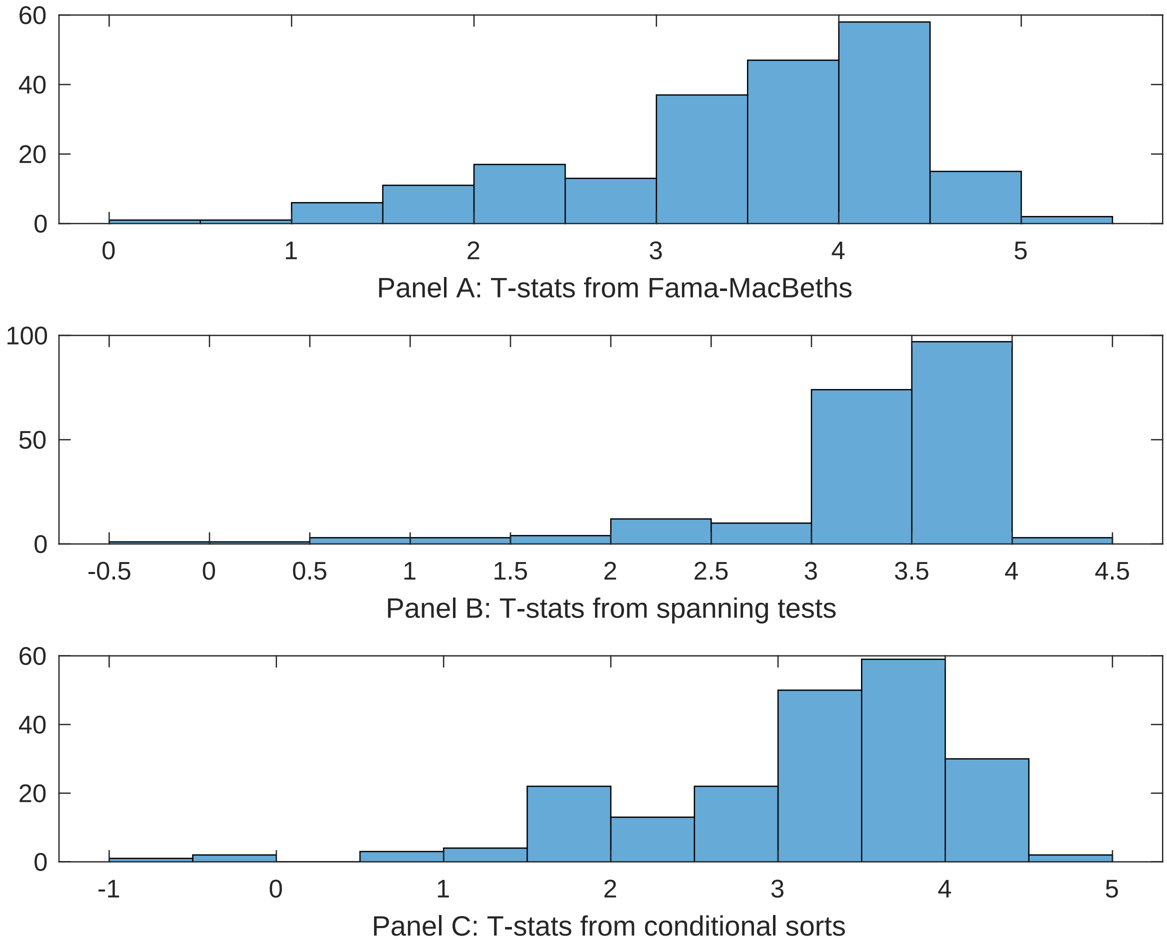


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TSI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.83	0.21	0.10	0.99	0.99	0.91	0.18
	[3.71]	[9.23]	[4.70]	[4.20]	[4.42]	[3.87]	[5.21]
TSI	0.41	0.44	0.43	0.57	0.51	0.19	0.31
	[3.47]	[3.68]	[3.89]	[4.55]	[4.44]	[1.32]	[1.81]
Anomaly 1	0.33						-0.14
	[4.95]						[-1.39]
Anomaly 2		0.61					0.43
		[5.61]					[3.16]
Anomaly 3			0.18				-0.30
			[2.38]				[-2.84]
Anomaly 4				0.91			0.15
				[4.72]			[4.49]
Anomaly 5					0.86		0.16
					[2.84]		[1.92]
Anomaly 6						0.36	0.29
						[4.90]	[3.67]
# months	727	720	727	667	727	655	631
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the TSI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TSI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the **?** five-factor model augmented with the **?** momentum factor. The six most closely related anomalies, X, are Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.44	0.42	0.38	0.41	0.34	0.30	0.34
	[4.40]	[4.15]	[3.78]	[4.08]	[3.44]	[2.94]	[3.56]
Anomaly 1	42.05						33.56
	[6.65]						[5.00]
Anomaly 2		18.67					-8.42
		[3.59]					[-1.30]
Anomaly 3			18.79				5.37
			[4.23]				[1.00]
Anomaly 4				-19.91			-20.55
				[-7.06]			[-7.55]
Anomaly 5					20.17		17.45
					[5.59]		[4.04]
Anomaly 6						15.62	10.11
						[6.23]	[4.19]
mkt	-1.03	-0.60	0.30	-0.45	-0.14	-1.40	-0.98
	[-0.43]	[-0.25]	[0.13]	[-0.19]	[-0.06]	[-0.59]	[-0.44]
smb	-16.71	-8.29	-6.41	-5.36	-5.72	-6.86	-20.82
	[-4.14]	[-2.18]	[-1.78]	[-1.55]	[-1.63]	[-1.96]	[-5.38]
hml	-23.72	-1.43	1.62	30.24	4.86	21.16	-11.16
	[-2.98]	[-0.19]	[0.26]	[6.25]	[0.92]	[4.60]	[-1.34]
rmw	-18.55	-27.48	-31.10	-20.69	-34.01	-20.97	-14.16
	[-3.83]	[-5.83]	[-6.54]	[-4.36]	[-7.12]	[-4.39]	[-2.86]
cma	-14.36	-8.91	-10.24	-8.36	-16.91	-11.77	-15.58
	[-2.13]	[-1.28]	[-1.49]	[-1.24]	[-2.47]	[-1.74]	[-2.34]
umd	3.09	4.50	7.62	-1.46	13.52	8.95	9.08
	[1.31]	[1.88]	[3.04]	[-0.59]	[4.70]	[3.63]	[3.09]
# months	728	732	728	722	728	728	722
$\bar{R}^2(\%)$	11	7	8	12	10	11	22

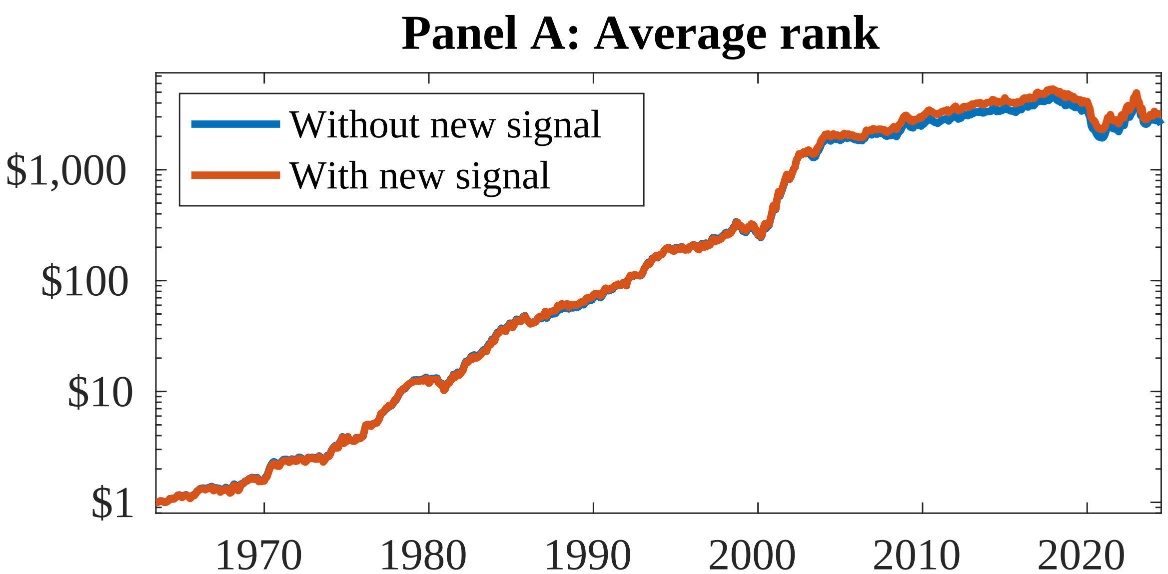


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following **?**. In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as TSI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.